Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/04/2006
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Aerodynamics and Chemical Kinetics of Premixed Flames at High Pressures		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Ground	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	note: renewed 4 yrs		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04AA09A		
Performance Goal No.:			
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Task Description:	The program consists of several parts, each of which consists of both experimental measurements and direct numerical simulation of microgravity flames: 1) Determination of the laminar flame speeds of gaseous fuels; 2) Determination of the corresponding Markstein lengths for these flames; 3) Determination of the extinction limits; 4) Investigation of Landau-Darrieus and diffusional-thermal flame surface instabilities; and 5) Recompilation of existing chemical kinetic mechanisms by using the high-pressure flame speed data as optimization points. The purpose of this research is to develop a fundamental understanding of premixed gas flame propagation, extinction and stability in high-pressure environments, where microgravity experiments will permit measurements undisturbed by convective effects. The chemical reaction mechanisms, which govern the various flame processes, are inherently nonlinear, and can cause non-monotonic variations of flame responses to increased pressure. Most practical combustion systems, especially internal combustion engines, operate at elevated pressures; while the fundamental information on flame chemistry were		

	obtained at substantially lower pressures. Contributions to the fundamental understanding of flame processes at elevated pressures will have direct relevance to advances in the development of efficient and low-emission internal combustion engines.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Contributions to the fundamental understanding of flame processes at elevated pressures will have direct relevance to advances in the development of efficient and low-emission internal combustion engines.
Task Progress:	We have succeeded in experimentally observing pulsating spiral instabilities over propagating flames at elevated pressures, as well as in theoretically formulating a model that describes such spirals. Furthermore, we have experimentally demonstrated that the emergence of these spiral waves follow closely the right stability boundary governing the onset of diffusional-thermal pulsating instability, provided the evaluation of the reduced Lewis number is based on locally extracted thermokinetic parameters, and that these spiral waves exhibit many of the same characteristics such as meandering often observed in other excitable media, most notably the Belousov-Zhabotinsky reaction. It is also of interest to note that these spiral waves seem to initiate over surface elements. When hydrodynamic cells are formed, the outwardly propagating spirals are eventually absorbed at the cell boundary. Furthermore, mixed boundary conditions representing the presence of both diffusion and convection are needed to theoretically generate the spirals. The phenomenon appears to be quite rich, meriting further study. While the present experiments were performed with rich hydrogen–air mixtures, the same phenomena of spiral flames could also exist for lean hydrocarbon–air mixtures with carbon numbers greater than two, such as propane, butane, and heptane, as the Lewis numbers for these mixtures of practical importance are also greater than unity. Indeed, computational studies of the one-dimensional planar propagation of lean heptane–air flames carried out in our lab have unambiguously shown the occurrence of pulsating instability prior to flame extinction. Furthermore, the sensitivity increases with leaner mixtures. The combustion and the associated unstable phenomena observed herein are hence of relevance to the utilization of petroleum fuels in internal combustion engines, especially for ultra-lean operations. Experimental laminar flame speeds were accurately measured for CO/H2/air and CO/H2/He mixtures at different equivalence ratios and mixing ratios up to 40 atmospheres using the constant-pressure spherical flame technique. This adds to our already substantial database of high-pressure laminar flame speeds and Markstein lengths, enabling us to be leaders in the field in this aspect of combustion research. Further, the reaction rate constant of CO+HO2 ®CO2 +OH was calculated based on ab initio theory and canonical transition state theory. A kinetic mechanism based on recently published and calculated rate constants was presented to model the measured laminar flame speeds as well as the experimental data from counterflow ignition, flow reactor, and shock tube experiments in the literature. The comparison between the modeling results and the laboratory measurements suggests that accurate thermochemical data and the elementary rate constants are crucial to the satisfactory performance of the reaction mechanism. Future work will continue to combine experimental, numerical, and theoretical work on high-pressure flame phenomena. We intend to increase the number of fuels in our database with higher hydrocarbon fuels, and use the experimental results to create and update comprehensive chemical kinetic mechanisms. Further, we will determine the effects on the flame speed of the cellular (increased) and pulsating (overall decrease) instabilities. Understanding the effects of these instabilities are key to improvement of internal combustion engines, as flame holding, extinction, knock, and chamber lifetime are all affected by fundamental flame instabilities not yet fully understood.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Jomaas G, Bechtold JK, Law CK. "Spiral waves in expanding hydrogen–air flames: Experiment and theory." Proceedings of the Combustion Institute, 2007 Jan;31(1):1039-46. http://dx.doi.org/10.1016/j.proci.2006.08.100 , Jan-2007
Articles in Peer-reviewed Journals	Sun HY, Yang SI, Jomaas G, Law CK. "High-pressure laminar flame speeds and kinetic modeling of carbon monoxide/hydrogen combustion." Proceedings of the Combustion Institute, 2007 Jan;31(1):439-46. http://dx.doi.org/10.1016/j.proci.2006.07.193 , Jan-2007
Articles in Peer-reviewed Journals	Law CK. "Propagation, structure, and limit phenomena of laminar flames at elevated pressures." Combustion Science and Technology, 2006;178(1-3):335-60. http://dx.doi.org/10.1080/00102200500290690 , Jan-2006